

Report
Of
LastKM — AI-Powered Backhaul Matching for Community Good

Submitted
To
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IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of
B.Tech CSE (AIML)



By
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CANDIDATE’S DECLARATION

I, ALEKH KUMAR, hereby declare that the internship report titled “LastKM — AI-Powered Backhaul Matching for Community Good” has been prepared by me to fulfil the requirement for the award of the degree of Bachelor of Technology in CSE. All references used in this report have been acknowledged. I affirm that this report is my own work and has not been submitted to any other institute for any degree or diploma requirement. I shall be solely responsible for any kind of copyright violation in this regard.

Signature: _____

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ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the AI for Sustainability Virtual Internship. I am thankful for the guidance, constructive feedback and advice that helped me understand how artificial intelligence can be applied to practical sustainability challenges.

I am grateful to IM1B, AICTE and IBM SkillsBuild for providing the internship and learning environment in which this project was developed. I also express my gratitude to my university and faculty for their academic support.

I express my gratitude to my parents for their blessings and encouragement.

Signature: _____

Student Name: ALEKH KUMAR

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INTERNSHIP COMPLETION CERTIFICATE

The certificate below is the IBM SkillsBuild completion certificate currently available for the 1M1B | AI for Sustainability Virtual Internship. The final internship certificate is to be received after the remaining project-submission/completion process.



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4. PROJECT DESCRIPTION

4.1 Introduction

LastKM is an AI-powered backhaul matching concept developed during the 1M1B AI for Sustainability Virtual Internship. The project explores how empty-return delivery trips can be used to support small community transport requirements without creating an additional dedicated trip.

The project addresses a situation in which a delivery vehicle may complete its outbound delivery and return with unused space. At the same time, students, individuals, shopkeepers and NGOs may have small, non-perishable items that need affordable local transport. LastKM proposes an AI-assisted matching workflow to connect compatible return trips with such requests.

The project is aligned with Sustainable Development Goal (SDG) 11, Sustainable Cities, and SDG 12, Responsible Consumption. The supplied project presentation describes the concept as a way to improve the utilization of routes that are already being driven.

The project includes a working browser prototype. The prototype allows a driver to register a route, available space and a time window, after which the matching engine checks route overlap, capacity fit and time-window compatibility for nearby community requests. [filecite\turn6file1\L75-L79](#) [filecite\turn6file4\L183-L214](#)

4.2 Organization Profile

The internship was offered by 1M1B in collaboration with AICTE (National Internship Portal) and co-certified by IBM SkillsBuild. The offer letter identifies the program as the “1M1B - AI for Sustainability Virtual Internship” and states that it was designed to develop industry-relevant AI skills and explore the intersection of AI and sustainable development. [filecite\turn6file0\L12-L21](#)

The internship included learning exposure to AI/ML concepts, large language models, IBM Granite models, retrieval-augmented generation (RAG) frameworks and agentic AI systems with IBM Labs. It also covered sustainability applications including climate action, clean energy, waste reduction, biodiversity and sustainable cities. [filecite\turn6file0\L22-L30](#)

The offer letter specifies a six-week internship period from 28 July 2026 to 28 August 2026 and requires participation in live sessions, guided learning modules, self-learning labs, digital badges and a final project submission containing a concept note, video and code repository. □filecite□turn6file0□L18-L21□

□filecite□turn6file0□L31-L38□

The internship is unpaid. The offer letter also states that certification depends on mandatory participation, weekly tasks and milestones, completion of the SkillsBuild learning plan and submission of the final project presentation.

□filecite□turn6file0□L42-L64□

4.3 Problem Statement

The central problem addressed by LastKM is: “How might we use AI to match empty-return delivery trips with local community transport needs, so that urban logistics can become more sustainable and inclusive?” The project presentation describes local delivery vehicles returning empty while nearby individuals, students, shopkeepers and NGOs may struggle to find affordable transport.

□filecite□turn6file8□L371-L377□

The proposed challenge is not simply to find any available vehicle. A useful match must satisfy multiple constraints at the same time: the pickup and drop locations should fit the route, the item should fit the available space, and the requested time should overlap with the driver's return window.

The project presentation gives an illustrative context figure that 35–40% of urban delivery return trips run completely empty. This report treats that figure as a project-presented estimate rather than an independently verified statistic.

□filecite□turn6file8□L373-L377□

4.4 Project Objectives

- Develop an AI-assisted concept for matching empty-return delivery capacity with local community transport requests.
- Use route, capacity and time-window compatibility as core matching conditions.
- Provide a simple map-style/community booking experience for pickup and drop requests.
- Suggest a transparent and fair backhaul price that can benefit both sender and driver.

- Provide live tracking and a pre-arrival alert within the proposed delivery workflow.
- Include trust and safety features such as package declaration and two-way ratings.
- Explore responsible use of AI through transparent matching, pricing and privacy considerations.
- Demonstrate the proposed workflow through a working browser-based prototype.

4.5 Scope of the Project

The current scope is focused on small-to-medium, non-perishable community deliveries. Examples presented by the project include books, clothes, stationery, dry goods and similar supplies.

The system considers a driver's start sector, end sector, available space and operating time window. Community requests contain pickup sector, drop sector, required capacity and a requested time window. The prototype then evaluates whether the request fits the driver's route and available resources.

The current concept does not treat cold-chain or temperature-sensitive transport as part of the ordinary matching workflow. Medicines, vaccines and perishable food would require a separate verified cold-chain fleet with appropriate certification and compliance before being considered.

The present work is a project prototype and proof of concept. It does not claim to be a production logistics service, certified transport network or complete payment/identity-verification platform.

4.6 Technologies and Tools Used

Technology / Tool	Use in the project
HTML	Browser-based structure for the LastKM prototype.
CSS	Interface layout, panels, cards, buttons and visual presentation.
JavaScript	Matching logic, request rendering, pricing calculation, tracking flow and interactive behaviour.
IBM Bob	Used during ideation/prototyping to draft the matching and pricing logic and workflow structure, as stated in the project presentation.
IBM SkillsBuild	Internship learning platform and co-certification context.

The prototype source demonstrates the browser-based implementation and explicitly provides driver route, capacity and time-window inputs. The interface then scans for

nearby community requests using route, capacity and time-window checks.

□filecite□turn6file4□L183-L214□

The project presentation also states that IBM Bob was used to draft the matching and pricing logic and generate the workflow structure during ideation and prototyping.

□filecite□turn6file6□L257-L277□

Although the internship offer letter lists IBM Granite, RAG and agentic AI among the learning areas, this report does not claim that these technologies were implemented in the LastKM prototype because the supplied prototype evidence does not establish their direct implementation. □filecite□turn6file0□L22-L30□

4.7 System Architecture

Layer	Function
User Input Layer	Driver enters route, available capacity and time window.
Community Request Layer	Stores request information such as pickup/drop sectors, capacity and time window.
Matching Layer	Evaluates route fit, capacity fit and time-window fit.
Pricing Layer	Generates a suggested backhaul price and exposes a price breakdown.
Delivery Workflow Layer	Handles acceptance, live tracking, ETA/pre-arrival notification, completion and rating.
Trust & Safety Layer	Supports package declaration, ratings, filtering, transparency and privacy controls.

The project presentation describes the prototype workflow as driver registration, AI scanning of nearby pickup/drop requests, match and fair-price suggestion, acceptance by both sides, live tracking with a pre-arrival alert, and delivery completion followed by ratings. □filecite□turn6file6□L257-L276□

4.8 Methodology

4.8.1 Problem Identification

The project begins by identifying unused capacity on return delivery trips and the parallel need for affordable local transport. The design challenge is framed around using an already-planned route rather than creating an additional trip.

4.8.2 Ideation

The proposed solution combines smart matching, map-based booking, live tracking, pre-arrival alerts and fair pricing. The presentation states that IBM Bob was used to draft matching and pricing logic and to structure the workflow during ideation and prototyping. □filecite□turn6file6□L257-L277□

4.8.3 Matching Process

The prototype asks the driver to provide a start sector, end sector, available space and operating time window. It then checks community requests for route overlap, capacity fit and time-window compatibility. □filecite□turn6file4□L183-L214□

4.8.4 Delivery Workflow

After a suitable match is suggested, both parties can accept the match. The proposed workflow then provides live tracking and a pre-arrival alert, followed by delivery completion and two-way rating. □filecite□turn6file6□L257-L276□

4.8.5 Responsible Design

The project includes transparency and safety considerations. The proposed system uses declared package information, two-way ratings, match filtering, transparent pricing and limited live-location sharing during the active delivery window.

4.9 Expected Outcomes

The expected outcome is a practical demonstration of how AI-assisted matching can improve the utilization of existing delivery capacity while addressing a community transport need.

- Better utilization of return delivery capacity.
- Potential additional earning opportunities for delivery drivers.
- Potentially lower transport costs for small community deliveries.
- More accessible transport support for students, individuals, shopkeepers and NGOs.
- Potential reduction in unnecessary empty vehicle travel and associated fuel use and emissions.
- A transparent matching workflow that explains route, capacity and time compatibility.
- An extensible prototype that can later be connected to real maps, payments, identity verification and operational logistics services.

The supplied project presentation describes the economic, social and environmental benefits as expected impact. Actual savings in fuel, emissions, time or income would require a real-world pilot and measured operational data; they are not claimed here as experimentally verified results.

4.10 Certificates of Completion and Other Communication Mail Proof

The offer letter confirms selection for the 1M1B - AI for Sustainability Virtual Internship, offered by 1M1B in collaboration with AICTE and co-certified by IBM SkillsBuild. It also states that an official internship certificate is provided upon successful completion of the internship requirements. □filecite□turn6file0□L12-L21□ □filecite□turn6file0□L35-L38□

The currently available IBM SkillsBuild certificate records completion of the “1M1B | AI for Sustainability Virtual Internship” learning plan (PLAN-56138075980A) by Alekh Kumar, with completion date 09 Aug 2026 (GMT). □filecite□turn6file3□L140-L147□

The final internship certificate is not included in this report because the project-submission and final-completion process is still ongoing. The final certificate can be appended to the report and repository after it is issued.

No additional communication-mail proof was supplied with the materials used to prepare this report. Therefore, no unsupported email content has been added.

5. BIBLIOGRAPHY / REFERENCES

1. 1M1B. AI for Sustainability Virtual Internship — Internship Offer Letter. Supplied project document, 2026.
2. 1M1B / IBM SkillsBuild. 1M1B | AI for Sustainability Virtual Internship — Completion Certificate. Completion date: 09 Aug 2026 (GMT).
3. LastKM. AI-Powered Backhaul Matching for Community Good — Project Presentation. Supplied internship project presentation, 2026.
4. LastKM. AI-Powered Backhaul Matching — Live Matching Engine Demo. Supplied HTML prototype, 2026.
5. IILM University, School of Computer Science and Engineering. Internship Report Format updated 26-27. Supplied report template.